

Navarettia fossalis Habitat Suitability and Availability in Future Climate Projection Scenarios in California, USA

Emily Ieva
School of Physical Sciences
University of California - Irvine

ABSTRACT:

The aim of this study is to examine the impacts that drought and temperature changes are projected to have on the *Navarettia fossalis* niche habitat availability. *Navarettia fossalis* is a California native plant species known to reside in seasonal vernal pools and other dynamic wetland ecosystems. This study was also meant to contribute data which can aid in conservation efforts for this plant's habitat and overall ecosystem. This study was carried out using ArcGIS Pro version 3.6, multidimensional raster datasets, conditional statements, and a variety of raster analysis tools. The results were presented in tables from excel and visual map products. Consistent decreases in suitable habitat area were observed in SSP5-8.5 and SSP3-7.0 scenarios, indicating drought and heat increases impacting *Navarettia fossalis* habitats. An overall decrease in *Navarettia fossalis* niche habitat availability was also observed, however an increase in area was measured in 2085 in the SSP1-2.6 scenario. This may have been due to the increased adaptability of dynamic ecosystem species.

Keywords: *Navarettia fossalis*, conservation, drought, heat, ArcGIS, habitat availability

INTRODUCTION:

Navarettia fossalis is a vascular plant species which can be found inhabiting vernal pool sites (Witham et. al., 2018; Spencer & Porter, 1997). Vernal pools are dynamic and biodiverse ecosystems which create seasonal wetland biomes from winter rainfall and storms. (Witham et. al., 2018). Around summertime and early autumn, these pools dry out and then fill up again as winter weather cycles back in (Witham et. al., 2018). This seasonality is what makes vernal pool ecosystems so dynamic (Witham et. al., 2018). Vernal pools are also known to cycle carbon and other important nutrients, which helps them to support a wide range of flora and fauna species (Kifner et. al., 2018) in a very niche ecosystem.

Because of the seasonal dynamic of vernal pools, the distribution of the surrounding vegetation is typically dependent on annual hydrological factors (Montrone et al., 2019). The amount of rainfall/precipitation and their timing both impact the vernal pools' existing flora (Montrone et al., 2019). Along with *Navarettia fossalis*, many other species from the *Navarettia* genus also inhabit these dynamic ecosystems such as *Navarettia leucocephala minima*, *Navarettia prostrata*, *Navarettia leuc pauciflora*, *Navarettia leuc plieantha*, and more (Spencer & Porter, 1997).

Vernal pool habitat areas are on the decline in California (Witham et. al., 2018), which impacts the population status of native Californian plants like *Navarettia fossalis*. In Los Angeles County, almost the entirety of vernal pool complexes have been lost (Fiesler & Drake, 2016). Where flora and fauna are concerned, freshwater and vernal pool ecosystems are estimated to have a greater decline in biodiversity than in other terrestrial ecosystems comparatively (Dixneuf et. al., 2021). In California, hydrological changes around vernal pool habitats impact the ecosystem, sending certain species into a population decline (Shin & Kneitel, 2019). With *Navarettia fossalis* already considered by the US Federal Government as a threatened species (Witham et. al., 2018), the need for conservation for these pools and habitats greatly increases.

As previously referenced, there have already been studies which aim to conserve and restore Vernal Pool ecosystems in California and other US states. These are studies which have indirectly conducted research on the ideal habitat and ecosystem for *Navarettia fossalis* (Witham et. al., 2018). However, subjects in this field are largely unstudied through the lens of the California native plant species which inhabit these declining ecosystems. This study aims to focus solely on the *Navarettia fossalis* species in California to understand how is the suitable habitat availability (i.e. vernal pools and seasonal wetlands) for *Navarettia fossalis* in California impacted by future IPCC climate projections of mean temperature and precipitation levels? And how can the results of this study aid in future conservation efforts for vernal pools and *Navarettia fossalis* in California?

MATERIALS & METHODS:

This study was carried out in ArcGIS Pro version 3.6, with the study area focused on the state of California in the United States. This study area was chosen because it is where the subject species, *Navarettia fossalis*, is native and endemic to. Datasets were collected from the Calflora website ("Observation Search", n.d.), CHELSA Bioclimate Projections (ArcGIS, 2025), PRISM 30 Year Normals, CALFIRE Vegetation Wildlife Habitat Relationship, and a 30 arc-second DEM of North America. Observation points found through the Calflora website were filtered to only include *Navarettia fossalis* points from all available sources specifically within the map area, and included both obscured records and surveys/checklists' data. From here, a csv file was downloaded to add into the map. This all took place in the Observation Search section of the Calflora website. CHELSA multidimensional raster data was first separated into individual layers by scenario (Vo, 2026), and then added into the maps accordingly. The specific variables that were drawn from the CHELSA datasets include SSP126 2055 (P1), SSP126 2085 (P2), SSP370 2055 (P1), SSP370 2085 (P2), SSP585 2055 (P1), and SSP585 2085 (P2). Other CHELSA Bioclimate Projection datasets used for this study include Annual Precipitation Bio12 (ArcGIS, 2025) and Annual Mean Temperature Bio1 (ArcGIS, 2025) as they . The rest of the datasets layers were downloaded and added to the maps. To prepare the map itself for spatial analysis, an output coordinate system was set to NAD 1983 (2011) California (Teale) Albers (Meters) to simplify future projections and appropriately display data.

The CALFIRE layer was the first to be prepped and processed. Once added to the map, values from the fveg22_1.gdb file were extracted into points and output as WHR_Extract. The Summarize tool was applied to the WHRNAME field in the output layer's attribute table to get the occurrence count. From this count, around 70% of the points were set to be included in the overall analysis, excluding points found in urban areas and working down from highest to lowest class in the Count_WHRNAME field. The ~70% of counts were then separated from the rest of the table through the Select By Attributes layer. The Lookup tool was then used, the original CALFIRE file layer was the input, WHRNAME was set as the Lookup Field, and the output was titled WHR. All CALFIRE layers made or added before WHR was created was removed from the contents. WHR was then filtered in accordance with the *Navarettia fossalis* observation points through the Extract Values to Points tool, making the output layer WHR_Filter. The Select by Attributes tool was then used on WHR_Filter to remove any null values in the RASTERVALU field. The output layer from this step was titled *Navarettia_fossalis_Filter*, and the RASTERVALU field was deleted from the new output layer.

The elevation layer was the next to be processed for analysis. The 30 arc-second DEM of North America was added onto the map and clipped to the study area and then projected to the map coordinate system. The Extract Values to Points tool was used next to determine the elevation range at which *Navarettia fossalis* had been observed. Using conditional statements and the Raster Calculator tool, the plant species elevation range layer was combined with the Wildlife Habitat Relationship (WHR) layer. The output layer from this was then reclassified to show 0 values as NODATA values, creating a final output layer called WHR_Elev.

A model for the current conditions for *Navarettia fossalis* was created using the prism_ppt_us_30s_2020_avg_30y.tif file. It was added into the map, clipped, and then projected to the map coordinate system, resulting in the output Precip_30y layer. The exact same measures were taken to the temperature mean PRISM.tif file, outputted as Temp_30y. These output files were analyzed with the *Navarettia_fossalis_filter* and WHR_Elev layers to draw out values that only pertain to the study plant's elevation and habitat variables. These layers were then reclassified to show 0 values as NODATA values, creating one output layer called Current. This output layer shows where suitable habitats currently are for *Navarettia fossalis*, with 0 indicating no habitat suitability and 1 indicating habitat suitability. This same model then needed to be created for each SSP scenario.

All of the separated raster layers, SSP126 2055 (P1), SSP126 2085 (P2), SSP370 2055 (P1), SSP370 2085 (P2), SSP585 2055 (P1), and SSP585 2085 (P2) were clipped to the study area and projected to the map coordinate systems. The Raster Calculator tool was then used on each individual layer to find habitat suitability under each scenario. They were also all reclassified and output as SSP126_P1, SSP126_P2, SSP370_P1, SSP370_P2, SSP585_P1, and SSP585_P2, all showing suitable or ecologically available habitat area based on the particular projections from each scenario. This area for each scenario was then calculated from cell size and counts in km². All of these values were put into a table in Microsoft Excel with rows being separated by scenario and columns by year (i.e. 2025/current, 2055 (P1's), 2085 (P2's)). From

here Figure 1 was made. The results of this study were presented in a line chart format, including all final variables in the same one to closely observe the differences between them.

RESULTS:

The results of the study show how much habitat area would be ecologically suitable for the study plant species, *Navarettia fossalis*. In the SSP1-2.6 scenario, suitable habitat area decreases from 75,502.43 km² in 2025 to 45,370 km² in 2055 projections, then slightly increases to 45,941.83 km² in 2085 (Figure 1). In the SSP3-7.0 scenario, habitat area decreases from 75,502.43 km² in 2025 to 36,580.81 km² in 2055 projections, then further declines to 15,515.53 in 2085 (Figure 1). Lastly, in the SSP5-8.5 scenario habitat area decreases from 75,502.43 km² in 2025, to 29,718.43 km² in 2055, and even further to 12,457.81 km² in 2085 projections (Figure 1). These three scenarios represent measurements increasing from the next in mean temperature and decreasing in precipitation, all relative to each other. In this study, SSP1-2.6 represents small increases in heat levels and small decreases in precipitation, SSP3-7.0 represents moderate increases in mean temperature, and moderate decreases in precipitation. SSP5-8.5 represents larger increases in ecological drought and heat levels (AchutaRao KM et. al., 2021). SSP5-8.5 scenario had the greatest amount of habitat area loss with an overall decrease of 63,044 km², then SSP3-7.0 with an overall decrease of 59,986 km² (AchutaRao KM et. al., 2021). SSP1-2.6 showed the least amount of change, with a maximum habitat area loss of 30,132 km² (AchutaRao KM et. al., 2021). The results show an overall decrease in suitable/ecologically available habitat area for *Navarettia fossalis*. Figures 2-5 are map products for visualizing habitat area.

DISCUSSION:

The aim of this study was to find how projected changes in drought and heat levels would impact the habitat availability for a native California plant species, *Navarettia fossalis*, and aid in conservation efforts for its typical wetland ecosystem. The results found a slight increase in habitat area during the 2085 timeframe of the SSP1-2.6 scenario, rising to 45,941.83 km² in 2085 from a previous decrease to 45,370 km² in 2055. Studies have shown that these behaviors of adaptation are dependent on their experienced microclimatic variables (Carnicer et. al., 2021), supporting the idea that species in dynamic ecosystems can respond more positively to climate change effects. Therefore, this small increase in habitat availability may indicate a higher level of adaptability against harsh environmental changes that only dynamic habitats would have. Vernal pool species are known to be more adaptive against microclimatic variables including drought and flooding (Witham et. al., 2018). Native species within these sorts of habitats are also known to have more efficient adaptation behaviors and climate change responses than non-native species in the same space (Witham et. al., 2018; Javornik & Collinge, 2016). These studies support the idea that the California native plant, *Navarettia fossalis*, could adapt positively to climate change effects up until a certain level of change, as this same potential adaptation was not found in the harsher two projection scenarios (SSP3-7.0 and SSP5-8.5).

The results from the more extreme precipitation and temperature scenarios, SSP3-7.0 and SSP5-8.5, showed the greatest declines in suitable habitat area throughout the study. The most extreme drought and heat scenario, SSP5-8.5 resulted in 63,044 km² of habitat loss. SSP3-7.0 followed close behind, losing 59,986 km² of suitable habitat for *Navarettia fossalis*. These results prove a great threat to the study plant species, showing that cases of extreme drought and heat negatively impact *Navarettia fossalis* habitats. Because *Navarettia fossalis* already being considered federally endangered (Witham et. al., 2018), the projected habitat loss from climate change impacts makes conservation efforts increasingly important for California's future landscape.

More work needs to be done in order to go about conserving California native species (i.e. *Navarettia fossalis*) and vernal pool ecosystems. It would be beneficial to continue adding mapped data for these subjects, as there are only three made previous to this study (Witham et. al., 2018). Adding more mapped data will provide the scientific community with valuable information. Simultaneously, using mapping techniques can be a productive way to understand and showcase the changes in season wetland habitat availability in California.

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APPENDIX:

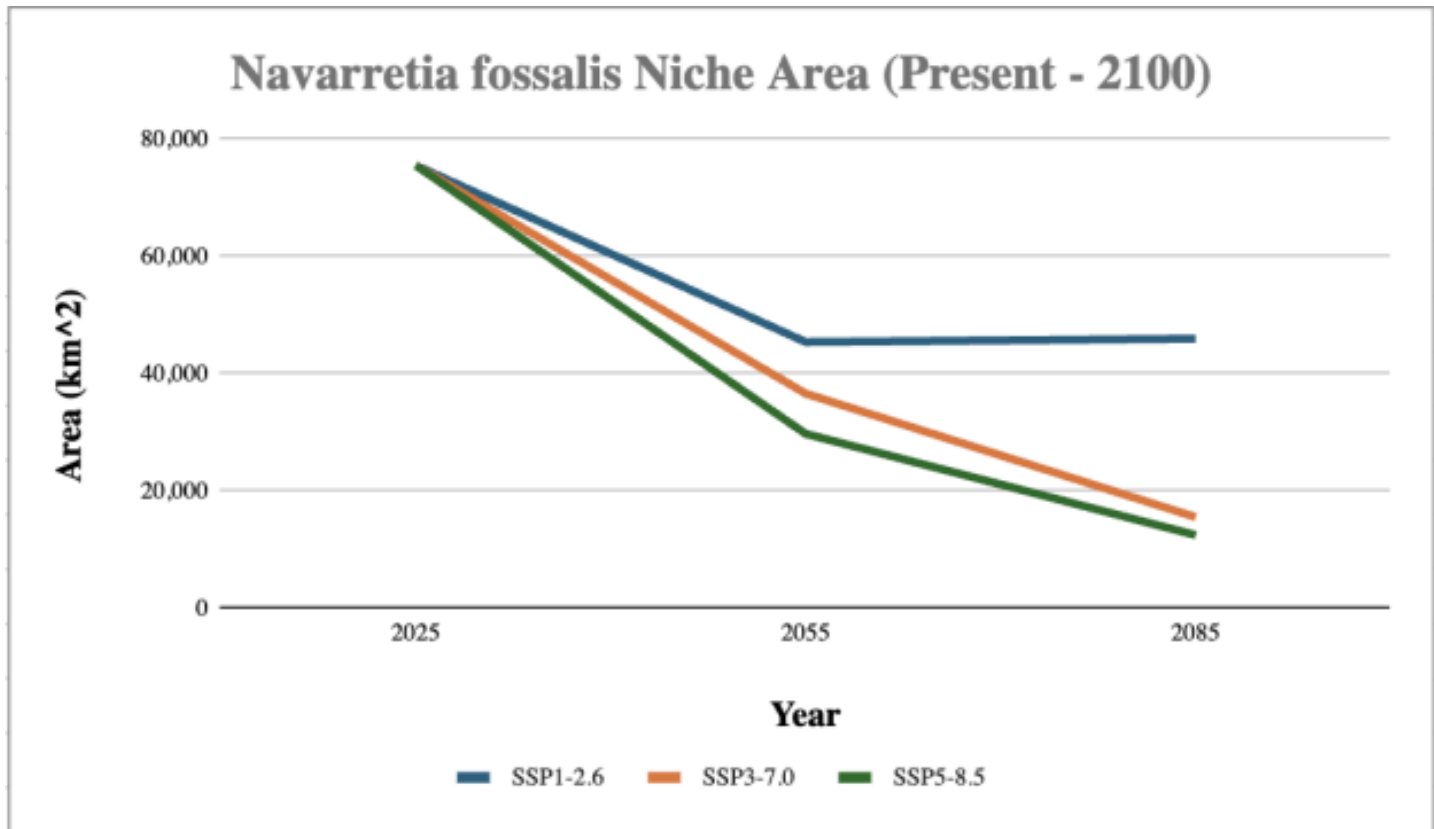


Figure 1. Shows the area of niche habitat availability for the *Navarretia fossalis* plant. Scenarios 3-7.0 and 5-8.5 show consistent decline in area for each year. Scenario 1-2.6 shows an initial decline in 2055, but then a very small increase in area in 2085.

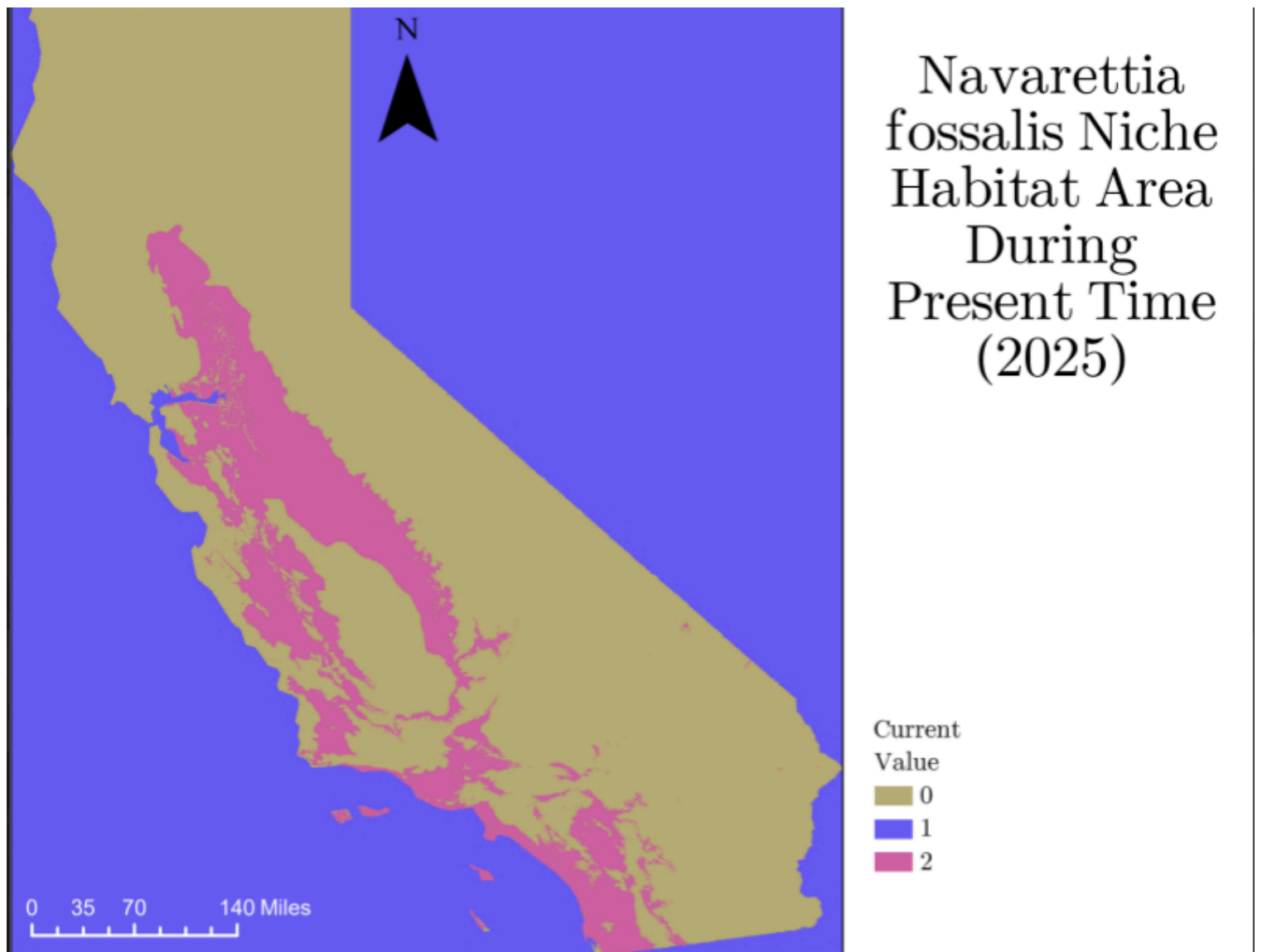


Figure 2. Shows the current (2025) Navarretia fossalis niche habitat area in pink. The tan shade indicates land that is unsuitable for the subject plant species and blue indicates the background so as to show just the state of California. Suitable habitat area can be found within the central valley, along the southern coast, among other scattered areas.

Navarettia fossalis Niche Habitat Area with SSP1-2.6 Scenario Drought and Heat Projections (2055 & 2085)

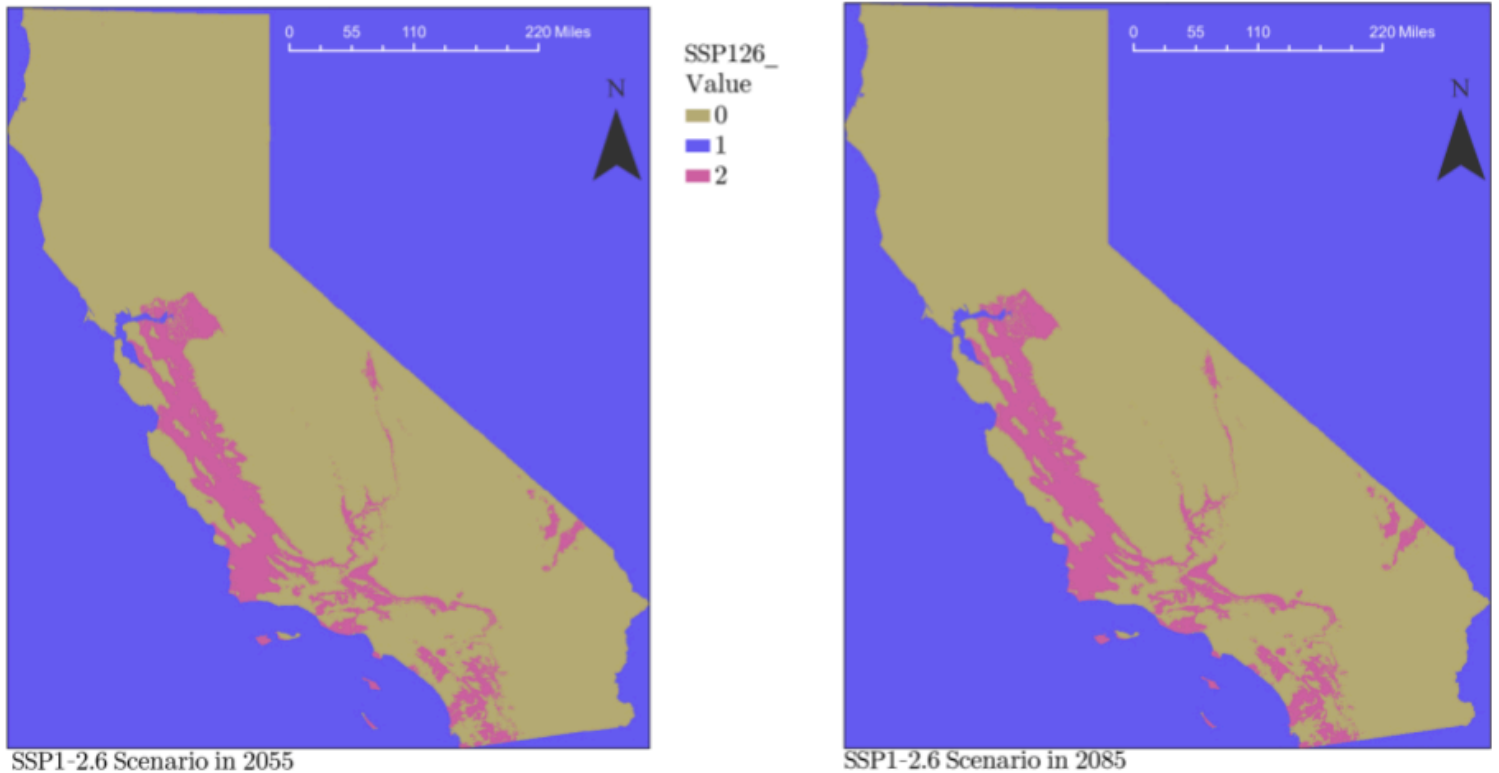


Figure 3. Shows *N. fossalis* niche habitat area according to the less harsh projection scenario (SSP1-2.6) during 2055 (left) and 2085 (right). Between these two maps there is in fact a small increase in area, but overall great decrease in area since the current model for 2025.

Navarettia fossalis Niche Habitat Area with SSP3-7.0 Scenario Drought and Heat Projections (2055 & 2085)

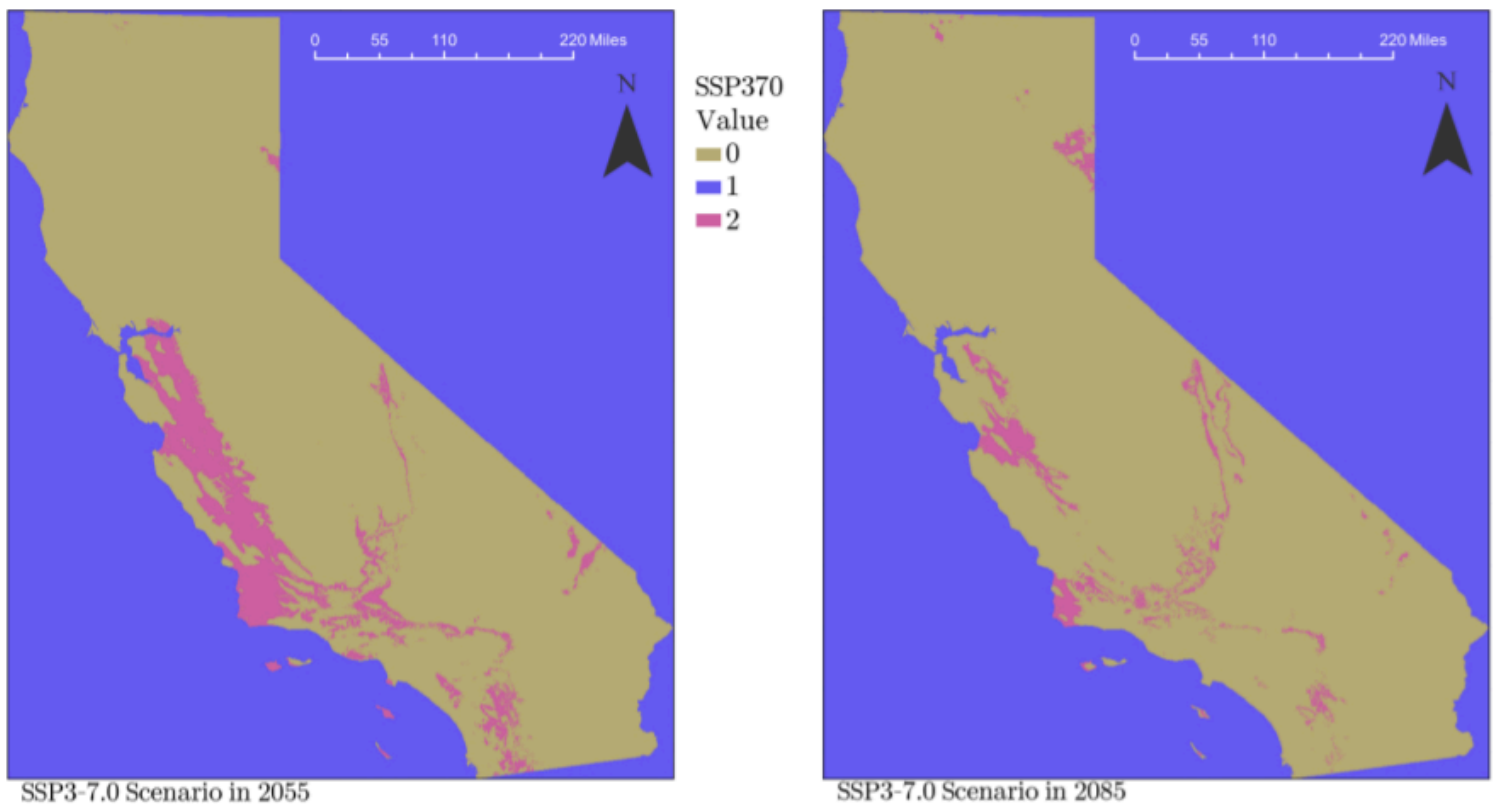


Figure 4. Shows *N. fossalis* niche habitat area according to the moderately harsh climate scenario (SSP3-7.0) for 2055 (left) and 2085 (right). Pink indicates suitable habitat area, blue gives the background to just show CA, and tan indicates unsuitability. These maps show a decrease in area throughout the timescale.

Navarettia fossalis Niche Habitat Area with SSP5-8.5 Scenario Drought and Heat Projections (2055 & 2085)

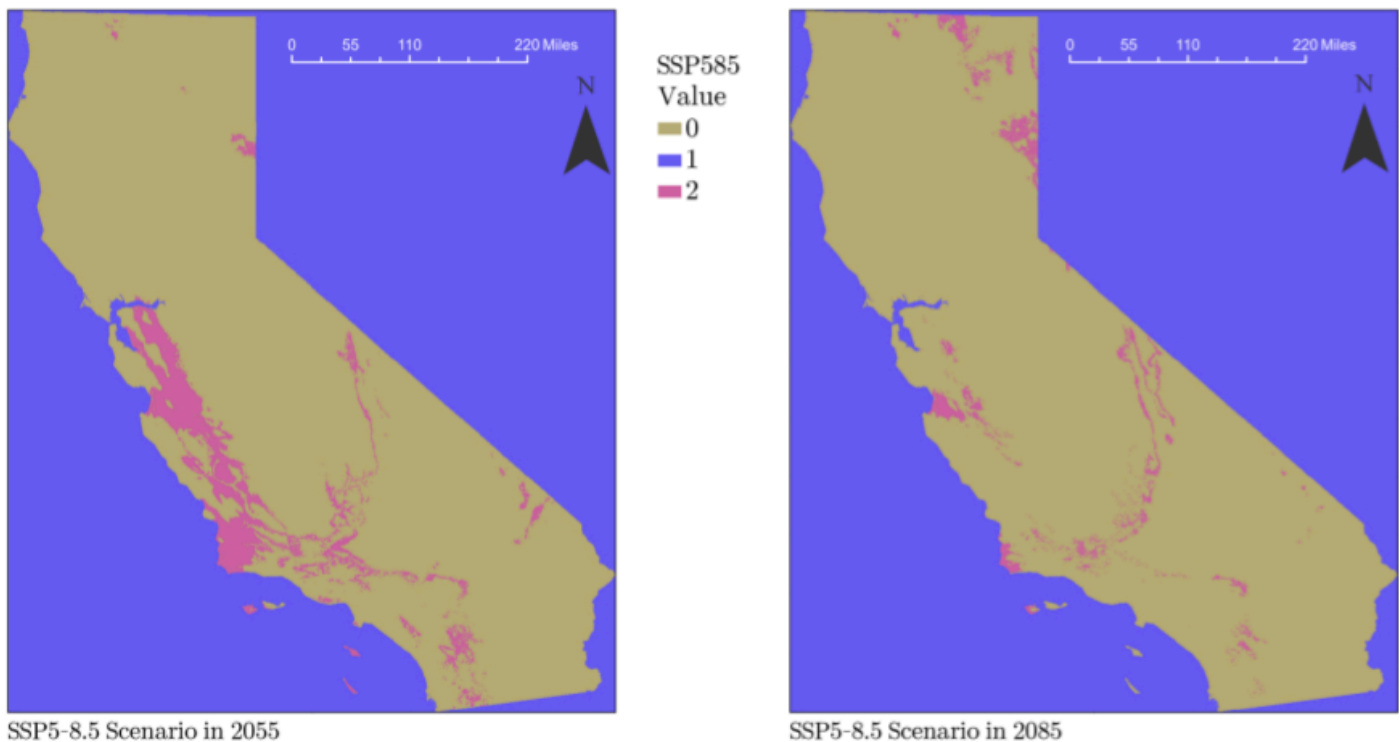


Figure 5. Shows *N. fossalis* niche habitat area according to the most harsh climate scenario of the study (SSP5-8.5) for 2055 (left) and 2085 (right). Pink indicates suitable habitat area, blue gives the background to just show CA, and tan indicates unsuitability. This scenario and these maps show the greatest amount of habitat area loss in the study.